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RECONNAISSANCE PALEOMAGNETIC INVESTIGATION OF THE OPHIOLITIC COMPLEX, THETFORD MINES, ASBESTOS, QUEBEC

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ABSTRACT

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A study was undertaken which shows that different metavolcanic units in the southern Appalachians of Quebec can be distinguished on the basis of their magnetic properties. At the same time, a preliminary paleomagnetic study of the rocks of this ophiolitic complex (peridotites, dunites, gabbros and pillow lavas) was undertaken. After AF demagnetization and unfolding of the geological formations, the paleopole position is 157°E, 06°N ($\alpha_{0.95} = 17°$). After thermal treatment, the paleopole position is 135°E, 18°N. Averaging the AF and thermally cleaned results, the paleopole position of this study is 146°E, 13°N. This pole position compares relatively well with four Cambrian paleomagnetic results of North America.

INTRODUCTION

This study is concerned with the magnetic properties and the paleomagnetic characteristics of the ophiolitic complex of the Asbestos and Thetford Mines in the Eastern Townships. The main objective of this preliminary work is to classify the different lithological units on the basis of their variable and characteristic magnetic properties, so that one can use these properties to distinguish similar lithological units (e.g. the different types of volcanics from the ophiolitic complex or the metalava from the Cambrian Caldwell Group) that are otherwise undistinguishable. This is a particularly useful approach in stratigraphic and structural problems and for detailed geological mapping of a specific region.

This preliminary study was also conducted to investigate the possibility of undertaking detailed paleomagnetic investigations of this ophiolitic complex in spite of alteration, complicated tectonic deformation, and metamorphism that the lithological units have undergone, in the hope of obtaining a better coverage of the geological time scale with paleomagnetic results of North American Phanerozoic rock units, particularly in the Cambrian and Ordovician periods for which the data are really scarce.

GEOLOGY

The ophiolitic complexes of the Asbestos and Thetford Mines are part of the Cambro-Ordovician rock sequences of the Appalachians of southern Quebec. They constitute a narrow belt some 300 km long, called the "Serpentine Zone" which extends from Vermont in the south to Maine in the north, passing through Orford, Asbestos, Thetford Mines, East Broughton, St. Favien de Panet, and St. Omer (Béland and Seguin, 1974) in the Province of Quebec.

The complete ophiolitic complex sequence consists of a basal amphibolite underlain by the Cambrian Caldwell Group (mainly greywackes and lava flows) and overlain by peridotite (asbestos-bearing harzburgite) of Alpine-type dunite and wehrlite, pyroxenite, gabbro and pillow lava. A cherty argillite is present on top of the pillow lava (volcanics and volcanic breccias). This Cambro-Ordovician sequence is overlain by the Early Ordovician St. Daniel Formation (phyllitic mélange) and finally by a Middle Ordovician flysch of the Magog Group.

The stratigraphy and structure of these different rock units have been investigated by many workers and in particular by Cooke (1937), Riordon (1953), St-Julien (1967), Lamarche (1972) and Laurent (1975). For the sake of clarity, we will include, in this section, a short petrographic description of the rock units investigated in this study, that is from bottom to top of the stratigraphic column of the ophiolitic complex: the peridotite, the dunite, the gabbro, the diabase and the metavolcanics.

The peridotite consists of harzburgite (80% olivine Fo_{90}, 18% orthopyroxene (enstatite), and 1—2% chromiferous spinel picotite) occasionally extensively serpentinized; some of the samples are composed of 50% or more of serpentine as in the case of the oriented sample 3-88-73, which is a serpentinite. The magnetite content increases with the amount of serpentinization. The dunite is composed of 85—90% olivine Fo_{90}, some clinopyroxene (diopside), 2—5% chromite and less than 1% magnetite. In the dunite as well as in the peridotite, the magnetite content increases with serpentinization. The gabbros are layered and their average composition is 40—60% plagioclase (An_{35-45}) and clinopyroxenes (diopside and augite); these clinopyroxenes are commonly amphibolized (hornblende). Minor amounts of magnetite are present. The pillow lavas contain olivine which is chloritized or silicified, pyroxene which is replaced by tremolitic actinolite, and calcic plagioclase which is replaced by albite, actinolite and small amounts of epidote. Veins of chlorite, epidote, quartz, calcite and disseminated pyrite and chalcopyrite as well as fine grains of magnetite are also present. Composite dike swarms and sills are numerous in the layered gabbros and pillow lavas horizons; they are composed of diabase, quartz diorite, spilitic keratophyre and granitic differentiates.

SAMPLING PROCEDURE AND MEASUREMENTS

In 1972, 73 and 74, some sixty selected oriented samples were collected by the hand sample method on different sites in the Thetford Mines—Asbestos

region. Fourty-one of these samples were judged acceptable for a paleomagnetic study. In this regional survey, efforts were made to select sites where there was an excellent exposure and where the structural geology was relatively simple and best understood. A geological map of the region investigated is shown in a paper by Laurent (1975). Great care was also taken to select fresh material to avoid alteration. Only regions of low-grade metamorphism were visited. The samples locations were chosen by reference to detailed geological maps published by the Department of Natural Resources of the Province of Quebec (1 mile to the inch). Except for the dunites and some peridotites, the formations are not anomalously magnetic and a magnetic compass could be used for the orientation of the rock samples.

The magnetic susceptibility was measured with a Soiltest MS-3 standardized resistance bridge having an accuracy of $5 \cdot 10^{-7}$ cgs emu. A two-probe fluxgate magnetometer (Förster Oerstedmeter) coupled to a custom built low speed (4 Hz) spinner magnetometer was used for measuring the intensity and direction of the NRM component. The alternating field (AF) demagnetization technique was used to remove unwanted secondary components and in particular the soft viscous magnetic components. For this purpose, an AC demagnetizer built at Université Laval and having a maximum peak-to-peak intensity of 1800 oersteds was used.

MAGNETIC PROPERTIES

The results relative to the magnetic susceptibility (K), the initial N.R.M. intensity (J) and the Koenigsberger ratio (Q) of the four most important rock types of the ophiolitic complex, that is, the pillow lavas, the gabbros, the dunites and the peridotites are presented in Table I.

The susceptibility of both the gabbros and the pillow lavas is of the order of $3 \cdot 10^{-5}$ cgs emu cm^{-3} and these two units cannot be distinguished on the basis of their susceptibility only. The susceptibility of the dunites averages $4 \cdot 10^{-3}$ cgs emu cm^{-3} and that of the peridotites $8 \cdot 10^{-4}$ cgs emu cm^{-3}. These two rock types have distinct susceptibility ranges. The high values obtained in the dunites are caused in part by the relatively high chromite ($FeCr_2O_4$) content. However, as the susceptibility of pure chromite is $3 \cdot 10^{-3}$ cgs emu cm^{-3} only (Seguin, 1971), some magnetite is also present.

In contrast to the susceptibility, the initial N.R.M. intensities of the pillow lavas and the gabbros are significantly different. The mean initial J-values of the gabbros ($5 \cdot 10^{-6}$ cgs emu cm^{-3}) are ten times larger than the ones of the pillow lavas ($5 \cdot 10^{-7}$ cgs emu cm^{-3}). If the initial J-values of the serpentinites are excluded from the peridotite group, the mean initial J-values of the dunites ($5.5 \cdot 10^{-3}$ cgs emu cm^{-3}) are about twice larger than the ones of the peridotite ($2.8 \cdot 10^{-3}$ cgs emu cm^{-3}). The large increase of initial J-values in the serpentinized peridotites is due for the most part to the increase in secondary Fe_3O_4 which occurred during the process of serpentinization. This secondary magnetite has a low coercivity and the accompanying secondary magnetization

TABLE I

Magnetic properties of different rock types from the ophiolitic complex

Rock type	Sample number	N	Susceptibility $(K) \cdot 10^{-5}$ emu/cm^3		N.R.M. intensity $(J) \cdot 10^{-6}$ emu/cm^3		Koenigsberger ratio (Q)	
			range	mean	range	mean	range	mean
Perido-	3-88-73 **	5	34—158	77	3650—26900	12450	6.15—103.5	47.49
tite	10-5-7-72	7	81—90	84	419—4330	3138	0.89—9.41	6.55
	6-09-7-74	7	77—105	90	1810—3045	2569	3.34—6.28	5.22
Mean values:			34—158	84	419—26900	6052	0.89—103.5	19.75
Dunite	5-10	6	236—273	254	3160—5580	4345	2.22—4.24	3.11
	3-78-74	4	493—679	569	4370—8260	6754	1.59—2.65	2.11
Mean values:			236—679	414	3160—8260	5549	1.59—4.24	2.60
Gabbro	3-78-73	6	2.7—3.9	3.1	1.6—7.4	5.1	0.07—1.22	0.45
	1-2-8-74	4	2.3—2.7	2.4	4.0—5.6	4.9	0.31—0.43	0.37
	2-14-8-73	5	2.3—2.8	2.5	5.1—6.7	5.8	0.36—0.52	0.42
Diabase Sill	18-28-7-72 *	4	208—232	219	611—1605	1022	0.49—1.24	0.84
Mean values:			2.3—3.9	2.7	1.6—7.4	5.3	0.07—1.22	0.41
Pillow	16-68-73	5	1.9—2.5	2.3	0.2—0.4	0.3	0.01—0.03	0.02
lava	16-18-73	7	3.0—3.6	3.2	0.5—1.2	0.7	0.03—0.07	0.04
Mean values:			1.9—3.6	2.7	0.2—1.2	0.5	0.01—0.07	0.03

* Mean value of Diabase Sill not included with the gabbro.
** Serpentinite.

is readily isolated by AC field demagnetization; at low-intensity AF demagnetization steps, the J-values decrease very rapidly.

As was expected, the Koenigsberger ratio (Q) of the gabbros (0.41) is at least ten times larger than that of the pillow lavas (0.03). The extremely low values of Q for the pillow lavas are caused by two coincident factors: the original low iron content of these oceanic basalts and the absence of secondary ferromagnetic minerals in these volcanics. The mean Q-values of the peridotites (5.9) are twice as large as the ones of the dunites (2.6) in spite of the fact that the reverse holds true for the initial N.R.M. intensity mean values. This is caused by the larger mean values of induced magnetization in the dunites by the earth's field at the sampling locality (0.58 Oe). It is difficult to conclude on the basis of the results of the mean Q-values alone whether the peridotites are more stable than the dunites in terms of paleomagnetism.

The data relative to the magnetic properties of the so-called "diabase sill" from Lac de l'Est do not fit in any of the ranges established for the four other rock types. The susceptibility of this sill lies between that of the dunites and the peridotites, its initial N.R.M. intensity lies between the one of the gabbros and the peridotites but closer to this last rock unit, and its Koenigsberger ratio

TABLE II

Statistical study of the paleopoles of the different rock types from the ophiolitic complex before AF demagnetization

Rock type	Sample number	N	Without fold test			With fold test		
			D_1 (°)	I_1 (°)	$\alpha_{0.95}$ (°)	D_2 (°)	I_2 (°)	$\alpha_{0.95}$ (°)
Peridotite	6-09-7-74	7	312	73	04	104	34	04
	10-5-7-72	7	331	73	06	295	14	05
	3-88-73	5	088	00	25	101	−62	11
Dunite	5-10	6	348	75	11	028	16	10
	3-78-74	4	024	−85	05	286	−18	04
Gabbro	3-78-73	6	053	73	16	108	−26	16
	1-2-8-74 *	4	023	26	07	085	01	06
	2-14-8-73	5	161	23	13	212	−57	15
Diabase Sill	18-28-7-72	4	212	75	11	143	−02	12
Pillow lava	16-68-73 *	5	118	06	17	117	−04	19
	16-18-73	7	024	18	21	041	16	18
Mean *:		60	036	70	26	073	−12	15
Paleopole position:			131°E	66°N	31	178°E	11°N	16

* Mixed polarities (?).

is located between the one of the gabbros and the dunites. This sill should then be classified as an ultrabasic (e.g. olivine diabase or dolerite).

Four lithological units of the ophiolitic complex are readily distinguishable on the basis of their differing magnetic properties. This new fact can be applied to solve specific stratigraphic, structural and other problems related to geological mapping in the ophiolite belt. A good knowledge of the magnetic properties is also of the utmost importance for the interpretation of aeromagnetic and ground magnetic data in the same region. In some instances, the proper interpretation of the magnetic data allows subsurface geological mapping or even surface or close-to-surface mapping in areas where overburden is present.

INITIAL N.R.M. ORIENTATIONS

The initial N.R.M. directions of the different lithological units of the ophiolitic complex investigated showed variable dispersion. Before AF demagnetization (see Table II), the mean direction of the N.R.M. component of all the

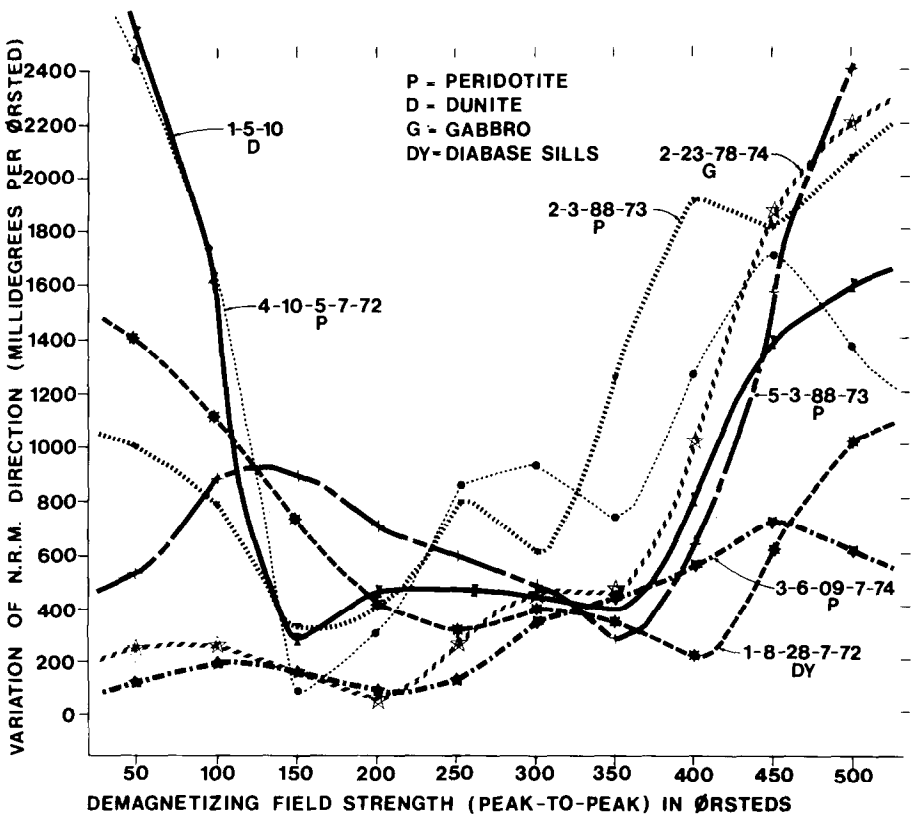


Fig. 1. Plot of the stability index of seven specimens of the geological formations investigated.

lithological units is 036°, 70° with a corresponding paleopole position of 131°E, 66°N (Table II) before geological unfolding. After unfolding, the N.R.M. mean orientation of the same rock units is 073°, −12° with a corresponding paleopole position of 178°E, 11°N. The main effect of the unfolding is to bring the longitude of the paleopole toward the west and the latitude from the north pole to the equator. The dispersion (Fisher, 1953) is smaller after unfolding. Polarity reversals were observed in samples 1-2-8-74 and 16-68-73 which are respectively a gabbro and pillow lavas collected from the same locality, that is Lac Coulombe. These preliminary results suggest the occurrence of a period of magnetic field reversal near the contact: gabbro–pillow lava, in the ophiolitic complex. Most commonly, the polarity is reversed, but this contact appears to be a preferred horizon for mixed polarities.

AF DEMAGNETIZATION

A minimum of two specimens from each site was demagnetized, in steps of 50 Oe, from 50 to 500 Oe in the absence of an ambient field. Guided by the

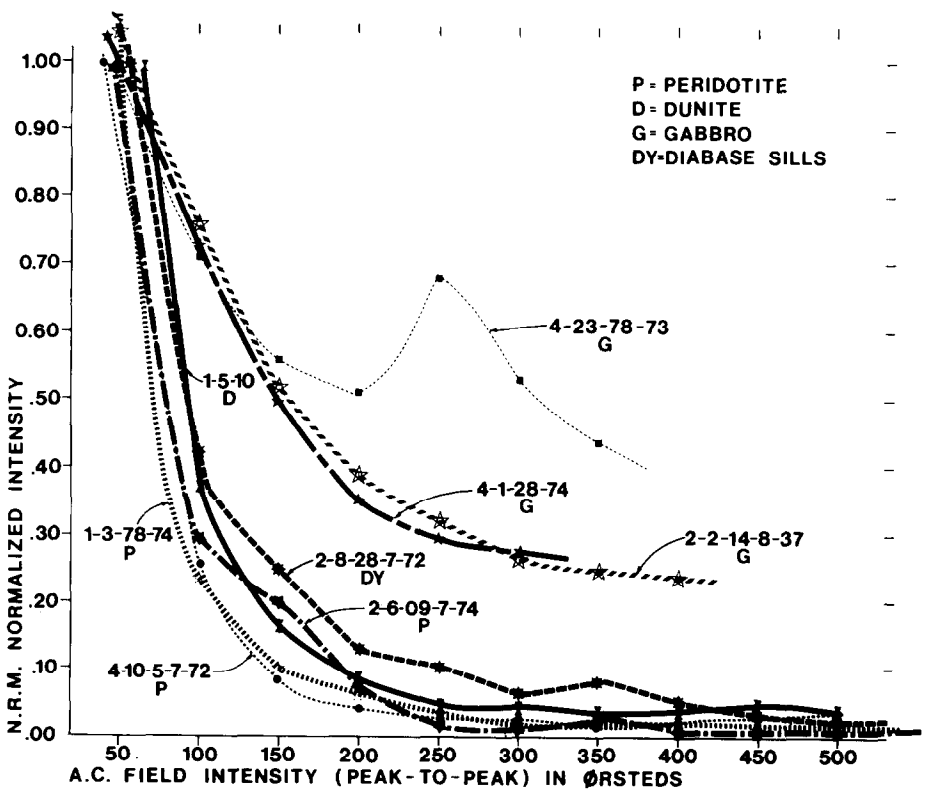


Fig. 2. Plot of normalized remanent intensity of seven typical specimens as a function of the demagnetizing field strength.

stepwise changes in remanence and intensity of the pilot specimens of a site, three intensities of AF demagnetization were selected using an "objective end point" stability method in which the remaining specimens from a site were AF demagnetized and their residual remanence remeasured. Pilot tests show that the AF intensity at which the primary thermoremanent component is most thoroughly isolated is about 150—400 Oe (Fig. 1). Normalized demagnetization intensity curves for some typical test specimens from a few sites are shown in Fig. 2. The magnetization directions of pilot specimens from three sites are presented in Fig. 3. After AF demagnetization, the dispersion within and between sites generally decreased and the site mean directions became grouped around a north—east axis with low inclination. The stable component remaining after AF cleaning is much smaller than the initial N.R.M. intensity; the average ratios of the mean site intensities, before and after clean-

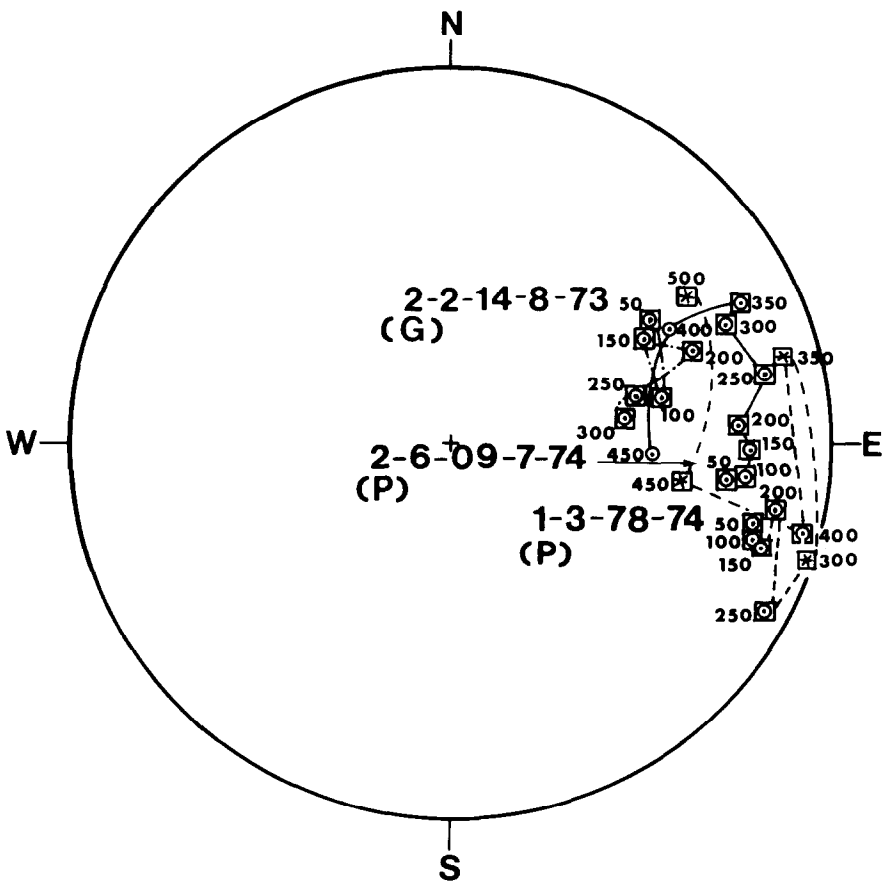


Fig. 3. Variation of the magnetization directions of three pilot specimens of the geological formations investigated as a function of the demagnetizing AF strength. Lambert equal-area plot. P = Peridotite, G = gabbro, ⊙ = upper hemisphere, * = lower hemisphere, □ = reversed polarity.

TABLE III

Statistical study of the paleopoles of the different rock types from the ophiolitic complex after AF demagnetization

Rock type	Sample number	N	AF intensity (oersteds)	Without fold test			With fold test		
				D_1 (°)	I_1 (°)	$\alpha_{0.95}$ (°)	D_2 (°)	I_2 (°)	$\alpha_{0.95}$ (°)
Peridotite	3-88-73	5	200	076	27	17	076	−27	16
	10-5-7-72	7	200	259	55	06	275	−08	07
	6-09-7-74	7	200	357	65	03	083	27	04
	Mean:	19		356	74	10	056	02	19
Paleopole position:			082°E	74°N		22	185°E	23°N	15
Dunite *	5-10	6	250	286	36	23	280	08	36
	3-78-74	4	250	011	−80	09	297	−12	09
	Mean:	10		323	−19	42	289	−03	14
Paleopole position:			304°E	63°S		45	326°E	12°N	21
Gabbro *	3-78-73		unstable						
	1-2-8-74	4	150	203	48	43	197	−43	44
	2-14-8-73	5	150	185	00	19	227	−27	17
	Mean:	9		192	22	27	221	−32	16
Paleopole position:			052°E	15°S		26	342°E	31°S	19
Diabase Sill	18-28-7-72	4	350	044	48	16	089	−02	06
Paleopole position:			164°E	48°N		12	163°E	00°	05
Pillow lava	16-68-73	5	100	328	18	27	105	30	14
	16-18-73	7	100	317	17	22	338	62	20
	Mean:	12		322	17	14	076	60	17
Paleopole position:			304°E	40°N		11	135°E	37°N	13
Paleopole position (all units)			353°E	57°N		20	157°E	06°N	17

* Reversed polarities.

ing at 200 Oe, is 0.23. The paleomagnetic significance and stability of the ophiolite suite are demonstrated by the departure of the site N.R.M. directions from the recent earth field direction and the evidence for reversal(s). After AF demagnetization and unfolding, the paleopole position is 157° E 06° N ($\alpha_{0,95}$ = 17°), see Table III.

THERMAL DEMAGNETIZATION

About 50% of the specimens were thermally demagnetized at temperatures of 100, 200, 300, 400, 500, 550 and 600°C. Roughly 60% of the samples were stable when subjected to thermal demagnetization. The magnetic unstability of the specimens was decided on the basis of the amount of variation of the intensity and orientation of the residual remanent vector, the change in polarity or all factors combined. Table IV shows examples of stability after thermal treatment and Table V examples of instability. After thermal demagneti-

TABLE IV

Examples of stable paleopole positions in the course of thermal demagnetization

Sample number	T (°C)	$J \cdot 10^{-6}$ emu/cm^3(cgs)	Paleopole position		Polarity
			longitude	latitude	
6-09-7-74	100	32.9	174	−06	R
	300	133	141	30	R
(P)	500	77.5	134	32	R
	550	82.4	158	−14	R
	600	35.4	117	37	R
3-78-73	100	2.28	131	03	R
	300	3.48	113	−04	N
(G)	500	3.06	75	10	N
	550	2.92	21	−69	N?
	600	1.49	159	−11	R
2-14-8-73	100	1.19	119	47	N
	300	2.32	137	33	N
(G)	500	2.18	113	34	N
	550	1.42	166	42	N
	600	0.23	93	−82	N?
3-78-74-1	100	1161	209	19	N
	300	473	121	31	N
(D)	500	348	120	33	N
	550	130	234	−74	N?
	600	253	144	−33	R
AVERAGE PALEOPOLE POSITION OF SEVEN THERMALLY DEMAGNETIZED SAMPLES: 135E, 18N					

R = reversed; N = normal.

TABLE V

Examples of unstable paleopole positions in the course of thermal demagnetization

Sample number	T (°C)	$J \cdot 10^{-6}$ emu/cm^3(cgs)	Paleopole position		Polarity
			longitude	latitude	
10-5-7-72	100	704	145	27	N
	300	537	249	10	N?
(P)	500	490	034	31	R?
	550	464	285	21	N
	600	496	025	51	R?
3-88-73	100	11.2	234	45	N
	300	125	025	21	R?
(P)	500	91.1	293	−17	R?
	550	55.1	063	22	N?
	600	176	094	22	N?
10-5	100	16.2	323	−37	R
	300	257	211	−42	R?
(D)	500	16.3	288	−10	R
	550	63.0	279	−05	R
	600	206	233	−07	R?
18-28-7-72	100	264	228	−51	R
	300	117	298	−34	R
(DIA)	500	83.2	188	46	N
	550	53.7	162	27	N
	600	38.2	316	−06	R
16-18-73	100	0.17	202	−27	N
	300	0.17	336	−15	R
(L)	500	0.13	107	−05	N
	550	0.39	348	−07	R
	600	2.02	061	21	N?

zation and unfolding, the paleopole position is 138°E, 18°N which is slightly at variance from the paleopole obtained after AF treatment.

AGE AND ORIGIN OF MAGNETIZATION

Absolute age determinations of the ophiolite suite are difficult to obtain. A provisional geochronological study yielded an age of 550 m.y. for the amphibolites of the ophiolitic complexes of the Appalachians of Quebec (Laurent and Vallerand, 1974). Few paleopole measurements are available for Cambrian rocks of North America. Three reliable Cambrian poles are listed in McElhinny (1973); their position is 140°E, 7°N ($\alpha_{0.95} = 22°$, $K = 31$) and, in all cases, the polarities are mixed or reversed. A reversed paleomagnetic pole at 164°E, 30°N was obtained by Rao (1975) for Lower Cambrian Bradore sandstones in Newfoundland. The paleopole position of this study is 146°E,

13°N when averaging AF and thermally cleaned results; this pole position compares reasonably well with the poles cited in the literature.

DISCUSSION OF THE RESULTS

This paleomagnetic study, even at this preliminary stage, is the first to be conducted in the ophiolitic complexes of southeastern Quebec. It is also one of the rare, if not unique, paleomagnetic studies incorporating ultramafic rocks such as peridotites and dunites. Hoye and Evans (1975) have recently shown that olivine-bearing rocks produce remanent magnetizations of paleomagnetic significance; this study indicates that their statement is quite correct. Even though the pillow lavas on top of the ophiolite sequence have a relatively low N.R.M. intensity, they can still be used for paleomagnetic studies since their magnetic signature is similar to those of the actual oceanic pillows (Seguin and Laurent, 1974). These preliminary observations suggest that detailed paleomagnetic studies can be undertaken on the differing lithological units of the ophiolite suite; such studies are now being pursued.

Both radiogenic and paleomagnetic data yield a Cambrian age for the ophiolitic complex. As the paleomagnetic results of this study are not significantly different from the ones of Black in 1964 (cited in McElhinny, 1973) and the one of Rao (1975), there is now less evidence to support the hypothesis of a large rotation of Newfoundland.

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